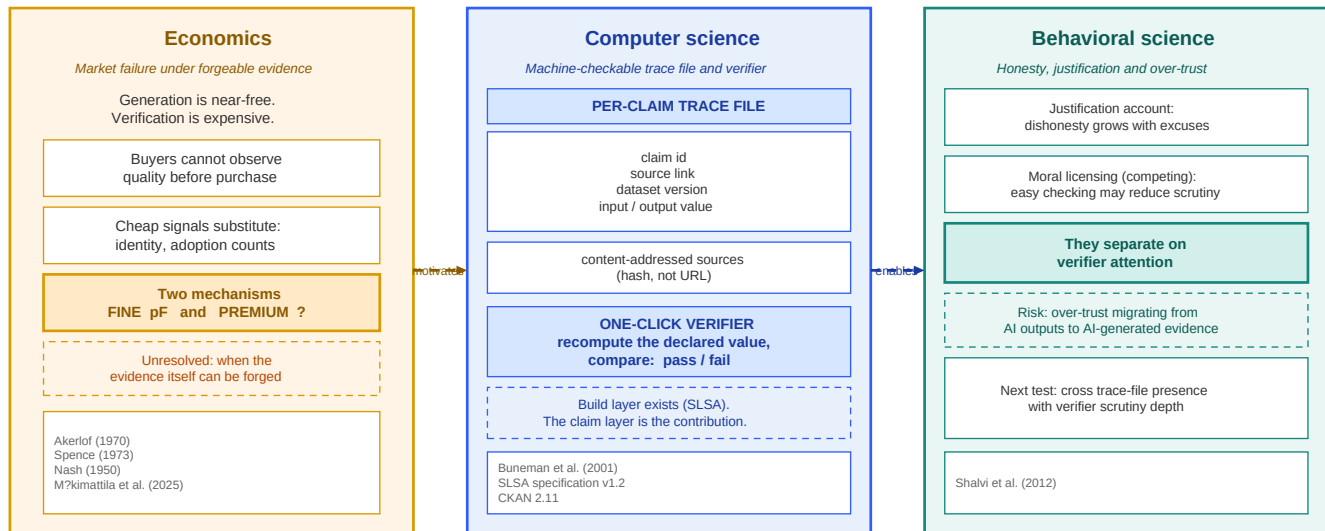


Honesty Through Two-Way Verification: A Credible Disclosure Infrastructure for the Era of Free Generation

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line, its mechanism, and the works cited for it. Solid boxes are established relations; dashed boxes and dashed arrows are the proposed synthesis. All values are simulated under the payoff law in the Colab notebook.

Figure 1: Prospective research design. Three disciplines, one strategic problem, with the works cited for each shown at the foot of its column. Free generation makes fabrication of the documents that carry information nearly costless, so the economic question shifts from what should be disclosed to which rules make honest disclosure stable [1, 6, 7, 10]. A machine-checkable per-claim trace file makes a declaration verifiable at near-zero cost, extending the build-layer provenance that already exists [2, 9]. Because effort-free checking may license inattention, the behavioral question is whether traceability shrinks the space for self-serving justification or invites over-trust [8]. Solid boxes are established relations; dashed boxes and dashed arrows are the proposed synthesis. Simulated outcomes, not observations of cognition or markets.

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Instructor: Luyao Zhang. **Assignment:** PS1 research proposal.

1 Three connected research questions

Generating a deliverable, and the documentation meant to vouch for it, is now nearly free, while independent verification remains costly. When anyone can fabricate the evidence, the question is not what to disclose but which rules make truthful disclosure stable. Stakeholders are plugin-marketplace developers, the platforms setting the rules, and the users and journalists who must act on a claim.

Three foundations frame the problem. Akerlof [1] shows unverifiable quality drives good sellers out of a market. Shalvi et al. [8] show dishonesty grows with available justifications. Buneman et

al. [2] define provenance as structured lineage, but the producer writes the record, so the trace itself can be forged.

Q1—Economics: Which mechanisms—a fine pF on detected fakes, or a market premium Δ for verifiable provenance—make the honest strategy a Nash equilibrium when trace files are free to generate and to fake?

Q2—Computation: What trace format and verifier make a per-claim declaration checkable at near-zero cost, without trusting the producer’s own record?

Q3—Behavior: Does traceability reduce dishonesty by shrinking the justification space, or does effortless checking produce moral licensing that increases it?

The disciplines are jointly necessary: a rule requiring trace files is empty unless a fake can be cheaply detected, and a rule that is enforceable may still fail if checking licenses inattention. Figure 1 connects the three.

2 Answer to Q1: incentives and intellectual lineage

Hypothesis. Two mechanisms can make honesty optimal, and either suffices. A penalty pF falls only on a detected fake; a premium Δ is earned only by a verifiable trace file. Honesty is the unique best response when $\Delta + pF$ exceeds the fake’s cost advantage.

Model. Developers choose an action $a_i \in \{\text{honest, none, fake}\}$ and verifier enrolment $v_i \in \{0, 1\}$; a platform commits to (m, p, F) for mandatory attachment, spot-check rate and fine, and the market sets a premium Δ for verifiable provenance. The platform sets rules, developers act, detected fakes pay F , and buyers pay Δ only for a trace they can check. Payoff is market benefit plus premium if honest, minus cost, plus stability if enrolled, minus expected fine if faking. A fake earns the same base market benefit as an honest deliverable, because buyers cannot observe truth before purchase, while saving cost. Information is incomplete. The solution concept is Nash equilibrium in pure strategies [7], with (m, p, F) exogenous; since the platform chooses the rules that developers best-respond to, this is a mechanism-design problem. I distinguish inherited quantities (Akerlof’s benchmark), simulated quantities (this payoff law) and hypothetical ones (real platforms).

Table 1: Strategy matrix with no rules ($p = F = 0$). Left: no premium, so faking wins. Right: a premium for verifiable provenance, so honesty wins without any enforcement.

		$\Delta = 0$	$\Delta = 30$
A	honest trace	135	165*
B	no trace	120	120
C	faked trace	160*	160

Without a premium, faking wins by 25—exactly the cost it avoids—and only a fine can deter it: $p^* = 0.5$ at $F = 50$. With a premium, honesty wins by being worth more and needs no enforcement. The mechanisms substitute: $\Delta = 15$ halves the required spot-check rate to 0.2, and $\Delta = 25$ removes the need to fine at all.

What the benchmark leaves unresolved. Akerlof shows unverifiable quality drives good sellers out of a market, but fixes buyers’ beliefs rather than asking how a costly signal forms them. Spence [10] supplies that logic: a premium exists only if the signal separates types. Neither says which rule restores honesty when the certification instrument is itself freely forgeable.

3 Answer to Q2: method and evaluation

Method. A browser-run 3×2 game on a Hugging Face Space, client-side HTML, CSS and JavaScript; the payoff law is documented in Appendix A. The algorithm enumerates the six cells and finds the best response.

Correction to the baseline. The first deployed version summed a cost, a benefit and a stability index. The notebook exposed the error: cost and benefit have opposite signs, so summing them scored honesty as better on both. The sweep returned $p^* = 0$, meaning faking was never attractive and the question could not be posed. The corrected law enters cost negatively and gives the fake the same market benefit as an honest deliverable. See Appendix B.

Baseline, modification, metric, check. Baseline is the status-quo cell (net 100); the modification is mandatory attachment plus p and F ; the metric is the threshold p^* at which honest-plus-verified is uniquely best. Five checks pass: with no rules the fake wins; with $p = 1, F = 100$ honesty is unique; the status quo is unchanged; enrolment is weakly dominant; there are six cells.

Result. At $F = 50$, $p^* = 0.5000$; the required p falls as F rises ($F = 100 \Rightarrow 0.25$; $F = 400 \Rightarrow 0.0625$), and below $F = 25$ no $p \leq 1$ suffices. All results are *simulated*.

Table 2: From laureate foundations to a new interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	Akerlof (1970): unverifiable quality collapses markets. Spence (1973): a credible signal needs single crossing. Mäkimattila et al. (2025): choice of certification is itself the signal.
PS1	A 3×2 game with a verified payoff matrix; two mechanisms located: a fine (pF) and a premium (Δ), with $\Delta + pF > 25$ required.
Next study	Endogenise Δ in buyer beliefs: test whether trace publication separates types, or pools and collapses. A test separating justification from licensing.
Longer term	A verifier a non-specialist can use, reporting which claims were checked, at what accuracy and what cost.
2056	Verifiable honesty that is competitively rewarded rather than merely enforced, and readable by any user.

4 Answer to Q3: behavior and competing explanations

Competing explanations. Justification: dishonesty grows with available justifications [8], so traceability should shrink that space. Moral licensing: a visible attestation lets producer and verifier treat the check as discharging responsibility, so the producer shades a claim the verifier no longer examines. They predict opposite effects from the same manipulation.

Discriminating evidence. They separate on verifier attention. Justification predicts less faking with scrutiny unchanged; licensing predicts reduced scrutiny and a fake rate falling less than the attachment rate rises. Measuring both producer choice and check depth discriminates; measuring the fake rate alone cannot.

Consequence. If licensing dominates, verification must not be a binary pass: it should report which claims were recomputed, and pF must be paired with something keeping scrutiny positive, since it deters only if p is real scrutiny.

Limits. A simulated table cannot establish human behavior. The next test is a short experiment crossing trace-file presence with check depth.

5 Advanced development and future directions

Foundation. Akerlof’s quality-uncertainty analysis [1], recognized by the 2001 Nobel Memorial Prize in Economic Sciences [11], establishes that unverifiable quality destroys markets and that institutions making quality observable restore trade. The enduring question is trust under asymmetric information. Spence [10] supplies the complementary logic: a costly credible signal separates quality, but only if the receiver can read it.

The boundary. Build-layer provenance [9] already exists. Unresolved is the claim layer: per-claim sources and input–output declarations, recomputed at near-zero cost, when the producer writes the record and can therefore forge it.

Where the premium must come from. Treating Δ as a constant merely relocates the assumption. A premium exists only if publishing a trace is informative about type, so that buyers update upward—the single-crossing condition that Spence [10] requires of any separating signal. If fakers can publish traces as cheaply as honest producers, the signal pools, Δ collapses to zero, and enforcement is again the only mechanism. Making Δ endogenous to buyer beliefs is therefore the next study, and the certification-design result of Mäkimattila et al. [6] is the natural instrument: there, what a sender reveals is not only the test outcome but the choice of which test to buy, and selection across tests is itself the signal. Its warning applies directly here—in that model the receiver obtains zero surplus, so a premium does not flow to honest producers automatically, and competition or rule design must be what directs it.

The verifier is built for machines. It recomputes a declared number and returns pass or fail, which a non-specialist cannot interpret. Letting an ordinary user judge a firm’s claim, or its data-handling practice, is a problem of cognitive rather than computational cost, where the objective is accurate enough and fast enough rather than perfect.

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, I aim to contribute a governed trust infrastructure for an era of free generation, in which verifiable provenance is competitively rewarded rather than merely enforced, and in which any ordinary user can check a firm’s claims and data-handling practices without technical training. Building on Akerlof’s quality-uncertainty foundation, Spence’s signalling logic and machine-checkable provenance, I will integrate economics, computer science and behavioral science so that near-zero-cost trace files, a verifier designed for non-specialists, and platform rules that make honest publication an equilibrium together let anyone inspect a claim in under a minute. Progress will be shown by an endogenous premium that separates honest from faking producers, a verifier whose accuracy and cost are documented, a discriminating behavioral test, and a rule design adopted on a live platform.

Open Science Statement

GitHub: <https://github.com/peterhe0616-sketch/PS1-Zhengjun>

Google Colab: <https://colab.research.google.com/drive/1RjQNsfFPo9dwrvxjKA>

Released: a browser-run simulator and a Colab notebook that re-derives the payoff table and runs the threshold sweep. The simulator is static client-side HTML/CSS/JavaScript with no backend, key or paid service; the notebook needs only Python. Inputs are synthetic: one payoff law read by both. No data are collected. Shortest run: open the notebook and choose *Run all*. Versions are in Appendix A.

Statement of Contribution to the UN’s SDGs

Hugging Face: <https://huggingface.co/spaces/Peterhe123/Trust>
Through an open-access game, this project supports **SDG 4—Quality Education** [12] by helping students and the public learn why honest disclosure requires cheap verification, by comparing the six rule-and-action combinations. No account or installation is needed. A feasible evaluation is a short pre/post question predicting which rule is stable. The resource released is the simulator; learning gains remain to be evaluated.

Submission milestones

This is v1, submitted September 13, 2026, 11:00 P.M. Beijing time (UTC+8). Class review of two assigned proposals: September 14. Author response to both reviews: September 16, 11:00 P.M. Reviewer follow-up: September 17. Revised submission v2 with its

updated sources, code snapshot and artifact links: September 20, 11:00 P.M. Appendix E records each reflection and revision; row 8 holds the feedback received so far and is marked pending for the peer reviews, which have not yet been received. No reviewer feedback is claimed in this version.

Author Notes

Acknowledgements

I thank Prof. Luyao Zhang, Program Chair, and Prof. Ken Rogerson, Program Discussant, of the *2056 Nobel and Turing Laureate Program Launch Workshop*, Duke Kunshan University, September 7, 2026; Longfei Jing and Yongzhuo Wu, my teammates in Workshop 6; and the rest of the COMSCI/ECON 206 Autumn 2026 Session 1 class. [Identify a specific contribution you actually used.] I am responsible for the proposal and remaining errors.

Individual contribution

I formed the core idea, posed the research question, and developed the one-click verifier, the lemons-market framing and the rule triple (m, p, F) . I verified the literature, played the game, and on 9 September found and corrected two errors in the payoff law (Appendix B). The photographs, the numbers and the final wording are mine.

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A Technical details and reproducibility

	Symbol	Definition
Notation.	a_i	Developer i ’s action: honest trace, no trace, or faked trace.
	v_i	Whether developer i enrolls in the platform verifier.
	m	Whether trace-file attachment is mandatory.
	p	Spot-check rate: probability a deliverable is inspected.
	F	Fine paid by a fake that is detected.
	pF	Expected fine on a faked trace file.
	Δ	Premium buyers pay only for a verifiable trace file.
	p^*	Smallest p making honest the unique best response.

Artifact. Trace-or-Fake Market Game, a 3×2 simulator. Space: https://huggingface.co/spaces/Peterhe123/Trust.Notebook:colab/trace_or_fake. Author: Zhengjun He. v1.0.0, 6 September 2026.

Payoff law. A developer chooses honest, none or fake, and whether to enrol in the verifier. Payoff is market benefit plus Δ if honest, minus cost, plus stability if enrolled, minus pF if faking. A fake earns the same base benefit as an honest deliverable, since buyers cannot observe truth before purchase, while saving cost.

Corrections, 9 September 2026. (i) The first law summed a cost, a benefit and a stability index. Cost and benefit have opposite signs, so summing them scored honesty as better on both before any rule applied, and the sweep returned $p^* = 0$: faking was never attractive, so the question could not be posed. Cost now enters negatively. (ii) Honest and faked traces were then paid identically, so honesty won only by avoiding a penalty. A premium Δ paid only for a verifiable trace file repairs this; the two mechanisms substitute, giving $\Delta + pF > 25$ against a fake advantage of 25. At $\Delta = 0$, $p^* = 0.5$ at $F = 50$; at $\Delta = 15$, $p^* = 0.2$; at $\Delta = 25$, no enforcement is needed.

Fresh-run record. Python 3.11, standard library only, deterministic, no seed needed. Six checks pass: no-rules fake wins; enforcement alone makes honesty unique; a premium alone ($\Delta = 30$) also does; status quo unchanged; enrolment weakly dominant; six cells. Abridged output:

```
no rules: A 135, B 120, C 160 -> best response
C_fake
threshold p* = 0.5000 (F = 50); F=100 -> 0.2500;
F=400 -> 0.0625
premium sweep (p=F=0): 0 -> C_fake; 25 -> tie;
30 -> A_honest
combined: delta=0 -> p*=0.50; delta=15 -> p*=0.20;
delta=25 -> no enforcement
```

Boundary. Static client-side HTML/CSS/JavaScript only: no runtime, backend, key or paid service. Hosting and the viewer’s browser still consume energy [5]. Semantic fieldsets, table captions and keyboard operability are present; a full screen-reader pass is unverified. The deployed Space should be updated to the corrected law.

A.1 AI-use disclosure

I completed all substantive work myself: the research question, the payoff law, the strategy matrix, the two corrections to that law, the derivation of the threshold $\Delta + pF > 25$, the computational checks, the game, the notebook and the argument of the paper. Every source was read and every number re-derived by me, and I played the game and ran the checker. A generative AI assistant (DeepSeek, model deepseek-v4-flash, accessed 6 September 2026) was used only for L^AT_EX typesetting and layout, and for brainstorming possible literature to look up. It did not devise the model, run any computation, or write the argument. All literature it suggested was independently located and verified before citation. Initial reasoning and handwritten reflection are Human-Only, and I remain responsible for every claim.

B Cumulative Intellectual Development
This proposal develops my Intellectual Statement II through the *2056 Nobel and Turing Laureate Program Launch Workshop*, September 7, 2026, Duke Kunshan University, IB 2050. Program Chair:

Prof. Luyao Zhang. Program Discussant: Prof. Ken Rogerson. My session: Workshop 6. My role: [assigned role]. My teammates: Longfei Jing and Yongzhuo Wu.

First change: the object of the question. My earlier claim was that I would build mechanisms letting agents cooperate under near-total unobservability. The September 4 evidence forced something narrower: at the Tencent Shanghai Office plugin quality is hard to observe before adoption, and at the Shanghai Science and Technology Museum the deep-residual-learning panel asks a general audience to trust institutional authority rather than checkable evidence. The marketplace is a market without institutional trust; the museum is institutional trust without a market. Workshop feedback sharpened the boundary—build-layer provenance already exists, so the contribution must sit at the claim layer—moving the proposal from a disclosure design to a rule design.

Second and third changes, 9 September 2026. The notebook sweep returned $p^* = 0$, showing the game summed cost with the wrong sign, so honesty won regardless of any rule and the game could not pose its own question. With that fixed, honest and faked traces were still paid identically, so honesty was a tax rather than an asset. Reading Mäkimattila et al. [6], where a sender’s *choice* among offered tests is itself informative, together with Spence’s single-crossing condition [10], prompted adding the premium Δ . The proposal can now show honesty wins because it is worth more, not merely because cheating is punished. What remains unresolved: Δ is still a parameter rather than a belief, so making it endogenous is the next study.

C Field and open-source inspiration

The *Joint Interdisciplinary Field Trip—Technology, Computation, Culture & Society in Action* took place on September 4, 2026.

Tencent Shanghai Office. Observation: WorkBuddy is Tencent’s AI agent platform, whose marketplace connects plugin developers with users. Interpretation: plugin quality is hard to observe before adoption, so buyers rely on cheap signals, and costly verification can produce a lemons market. Photograph: *WorkBuddy agent-plugin marketplace*, by the author, 4 September 2026.

Shanghai Science and Technology Museum. Observation: the 2023 Mathematics and Computer Science Prize panel [4] presents deep residual learning through simplified diagrams. Interpretation: trust here rests on institutional authority, not on independently checkable evidence. Photograph: *2023 Mathematics and Computer Science Prize panel*, by the author, 4 September 2026.

Two open-source tools. (i) SLSA [9], <https://slsa.dev/>, v1.2, Apache-2.0: generates non-forgable build provenance, so a verifier can check which commit built which artifact, but certifies the build chain only, not whether the artifact does what its description claims. (ii) CKAN [3], <https://ckan.org/>, 2.11, AGPL-3.0: an open-data catalog with standardised metadata and public APIs that stores whatever publishers upload and never checks whether a dataset matches its description. Consequence: SLSA supplies the build layer at near-zero cost and CKAN the public substrate, so my contribution is the claim layer both ignore.

D Review response and revision record

No peer review has been received as of v1. In-class review is September 14, 2026; no reviewer feedback is claimed. This section will be completed with the author response and the v2 revision.

E Structured abstract and cumulative proposal map

Table 3: Appendix E structured abstract; the eight-row proposal structure.

Proposal element	Current statement
Background	Build-layer traceability already exists and generation is near-free, while unchecked quality uncertainty collapses markets. Foundations: Akerlof (1970); Buneman et al. (2001).
Motivation and application	Untested: which mechanisms make honest trace files an equilibrium when they are free to generate and to fake. Scenario: a plugin marketplace setting the rules, with users and journalists as verifiers; it scales to public claims.
Three connected questions	Economics: does a fine pF or a premium Δ make honesty a Nash equilibrium? Computation: what format and verifier make a per-claim declaration checkable at near-zero cost? Behavior: does traceability shrink the justification space, or does easy checking license less scrutiny? Integrated: which rules make honest disclosure stable when the evidence can be forged?
Method and model assumptions	Developers choose honest, none or fake, plus enrolment; the platform commits to (m, p, F) and the market sets Δ . Information is incomplete. Payoff is benefit plus Δ if honest, minus cost, plus stability if enrolled, minus pF if faking; a fake earns the same base benefit as an honest deliverable. Baseline nets 120; the computation is a 3×2 enumeration plus sweeps of p and Δ .
Results	Verified: with no rules and $\Delta = 0$ the fake wins (160 vs 135); a premium alone restores honesty, and without one honesty requires $pF > 25$ ($p^* = 0.5000$ at $F = 50$). The two substitute: $\Delta = 15$ gives $p^* = 0.20$; $\Delta = 25$ needs no enforcement. Six checks pass. Evidence type: simulated, from one shared payoff law. Two corrections; see Appendix B.
Intellectual merits	Checkability becomes an institutional variable rather than a technical property, extending lemons theory to claims under free generation, where the certification instrument is itself forgeable. Separating enforcement from reward and showing they are substitutes reframes the design problem from policing to pricing.
Practical impacts	Private: lower screening costs on plugin and agent platforms. Public: auditable claims in museums, ministries and data portals. The SDG 4 tool lets students and the public learn why honest disclosure needs cheap verification, assessable by a short pre/post prediction question. The simulator is released; learning gains remain unevaluated.
Feedback received and revision made	September 7 workshop feedback established that build-layer provenance already exists, moving the contribution to the claim layer and revising the 2056 statement from cooperation under unobservability to a governed trust infrastructure. September 9, two dated changes: the sweep returned $p^* = 0$, exposing a sign error; and reading Mäkimattila et al. [6] showed honesty was modelled as a cost rather than an asset, so a premium Δ was added. Peer reviews pending; none claimed.